



Vidyavardhini's College of Engineering and Technology
Department of Artificial Intelligence & Data Science

AY: 2024-25

Class:	SE	Semester:	III
Course Code:	CSL304	Course Name:	OBJECT-ORIENTED PROGRAMMING IN JAVA

Name of Student:	SHRUTI GAUHANDRA
Roll No. :	15
Assignment No.:	02
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	3
Legibility	2	2
Total	10	10

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : MS. NEHA RAUT

Signature :

Date :

Q1. Write a Java class named Triangle that includes methods to:

1. Accept the base and height of the triangle
2. Calculate the area of the triangle.
3. Display the calculated area.

→ Code:

```
import java.io.*;  
import java.util.*;
```

```
class Triangle  
{
```

```
    private double base;  
    private double height;  
    private double area;
```

```
    public void acceptinput()  
    {
```

```
        Scanner scanner = new Scanner(System.in);  
        System.out.println("Enter the base of triangle:");  
        base = scanner.nextDouble();
```

```
        System.out.println("Enter the height of  
        triangle:");  
        height = scanner.nextDouble();
```

```
    }
```

```
    public void area()  
    {
```

```
        area = 0.5 * base * height;
```

FOR EDUCATIONAL USE


```

    }
    public void displayarea()
    {
        System.out.println("The area of the
                           triangle is : "+area);
    }
}

class Main
{
    public static void main (String args[])
    {
        Triangle triangle = new Triangle();
        triangle.acceptinput();
        triangle.area();
        triangle.displayarea();
    }
}

```

Output:

Enter the base of triangle : 6

Enter the height of triangle : 12

The area of the triangle is : 36

Q2. Create a class Rectangle with a default constructor that sets the length and width to 1, and a parameterized constructor that accepts length and width as parameters. Implement methods to calculate and display the area.
→ code.

```
import java.io.*;  
import java.util.*;
```

```
public class Rectangle {
```

```
    private double length;  
    private double width;
```

```
    public Rectangle() {
```

```
        this.length = 1;  
        this.width = 1;
```

```
    }
```

```
    public Rectangle(double length, double width) {
```

```
        this.length = length;
```

```
        this.do
```

```
        this.width = width;
```

```
    }
```



```
public double area()  
{  
    return length * width;  
}
```

```
public void displayarea()  
{  
    double area = area();  
    System.out.println("The area of Rectangle  
is: ", +area.);  
}
```

```
public static void main(String args[])  
{  
    Rectangle rec = new Rectangle();  
    Scanner sc = new Scanner(System.in);  
    System.out.println("Enter the length");  
    length = sc.nextDouble();  
    System.out.println("Enter width");  
    width = sc.nextDouble();  
    Rectangle rect = new Rectangle(length, width);  
    rec.displayarea();  
}
```

Output :

Enter length = 2

Enter width = 6

The Area of Rectangle is : 12.

Q3. Create a class Student that has constructor for initializing student details such as name, roll number and marks. Overload the constructor to allow creation of a student with just name and roll number. Implement a method to display student's detail.

→ code:

```
import java.io.*;
import java.util.*;
```

```
public class Student
{
```

```
    private String name;
    private int rollnumber;
    private double marks;
```

```
    public Student(Student name,
```

```
    public Student(String name, int rollnumber,
        double marks)
    {
```

```
        this.name = name;
```

```
        this.rollnumber = rollnumber;
```

```
        this.marks = marks;
```

```
    }
```

```
    public Student(String name, int rollnumber)
    {
```

```
        this.name = name;
```

```
        this.rollnumber = rollnumber;
```

```
        this.marks = 0.0;
```



```

    }
    public void displayDetails()
    {
        System.out.println("Student Name: " + name);
        System.out.println("Roll Number: " + rollnumber);
        if (marks == 0.0)
        {
            System.out.println("Marks: Not Available");
        }
        else
        {
            System.out.println("Marks: " + marks);
        }
    }
}
public static void main (String args[])
{
    Student student1 = new Student("Alice, 101, 85.5");
    System.out.println("Details of student 1:");
    student1.displayDetails();

    Student student2 = new Student("Bob, 102");
    System.out.println("Details of student 2:");
    student2.displayDetails();
}
}

```

Output:

Details of student 1:
Student Name: Alice

Roll Number: 101

Marks: 85.5

Details of student 2:

Student Name: Bob

Roll Number: 102

Marks: Not Available.